

Machine Learning and its Applications

Today's goals

Module 1

Machine Learning Fundamentals I: Basic Concepts

Welcome

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Welcome to the specialization - Applied AI for Engineers and Scientists: Foundations

Video • 6 min

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Specialization and Course Structure

Reading • 10 min

●

Specialization Sample Certificate

Reading • 10 min

●

Stay connected with UofG Online

Reading • 5 min

Module 7 Introduction

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Introduction to Module 7's Study

Video • 1 min

●

How to Access MATLAB Online

Reading • 5 min

●

Machine Learning in Engineering Practice: Tool, Replacement, or Decision Aid?

Reading • 5 min

Module 7-1 Machine Learning Fundamentals

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Machine Learning Fundamentals: What is Machine Learning

Video • 6 min

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Machine Learning Fundamentals: Fundamental Concepts in Machine Learning (1)

Video • 6 min

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Machine Learning Fundamentals: Fundamental Concepts in Machine Learning (2)

Video • 8 min

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Materials for Machine Learning Fundamentals

Reading • 10 min

●

From Explicit Rules to Learned Models: What Actually Changes?

Reading • 5 min

●

Module 1 Learning Coach

Dialogue • 5 min

Module 7-2 Mapping Inputs to Outputs

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Mapping Inputs to Outputs: Data Representation

Video • 10 min

●

Mapping Inputs to Outputs: Parametric ML Model

Video • 7 min

●

Mapping Inputs to Outputs: Non-Parametric ML Model

Video • 7 min

●

Mapping Inputs to Outputs: Evaluate Output

Video • 2 min

●

Materials for Mapping Inputs to Outputs

Reading • 10 min

●

How Data Representation and Model Choice Shape What a Model Can Learn

Reading • 5 min

Module 7-3 MATLAB Implementation

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MATLAB Implementation: Simple Linear Regression and KNN

Video • 6 min

●

Materials about MATLAB Implementation

Reading • 10 min

●

From Concept to Code: What the Model Is Actually Doing

Reading • 5 min

Module 7 Quiz

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Module 7 Quiz

Practice Assignment • 30 min

Module 7 Assignments

●

Module 7 Assignment 1

Graded App Item • 30 min

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Module 7 Assignment 2

Graded App Item • 30 min

●

CSV File for Module 7 Assignment

Reading • 1 min

●

Module 7 Recap

Reading • 5 min

Module 2

Machine Learning Fundamentals II: Model Training and Evaluation

Module 3

Data Preparation

Module 4

Support Vector Machines

Module 5

Artificial Neural Networks

From Explicit Rules to Learned Models: What Actually Changes?

0:00 / 0:36

2x

Traditional programming and machine learning differ fundamentally in *where the logic lives*.

Using **one concrete example** from science or engineering, explain:

- what the inputs are,
- what the outputs are, and
- why writing explicit rules would be difficult or impractical.

For your example:

- Is the task classification or regression?
- What would count as a feature and what would be the label?

This course focuses mainly on **supervised learning**.

- In your opinion, what is the **main risk** if training data is poorly defined or poorly split when using supervised learning in engineering applications?

Go to next item

✓ Completed

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